# ASSIGNMENT

## KINEMATIC AND DYNAMIC ANALYSIS OF A LINKAGE MECHANISM

For the six-bar linkage mechanism shown below, determine:

$l_1 = OE =$

$l_2 = OA =$

$l_3 = AB =$

$l_4 = BE =$

$l_5 = CD =$

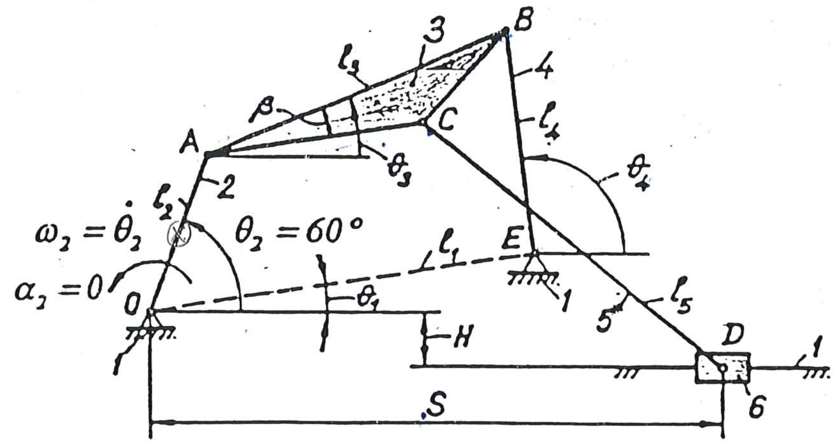
$AC =$

$H =$

$\beta =$

$\theta_1 =$

$n_2 =$




1. All possible positions of links and joints by graphical position analysis. Draw to scale all positions of joints for 16 subsequent positions of link 2 and determine the limits of motion where appropriate. Identify and outline the paths of each moving joint.
2. Linear and angular velocities by graphical velocity analysis for the given position $\theta_2 = 60°$ of the mechanism, as shown above. Draw to scale the velocity vector diagram encompassing all linear velocities to scale, and present the results in a tabular form.
3. Linear and angular accelerations by graphical acceleration analysis for the given position $\theta_2 = 60°$ of the mechanism, as shown above. Draw to scale the acceleration vector diagram encompassing all linear accelerations to scale, and present the results in a tabular form.
4. All instantaneous centres of velocity for the given mechanism using Kennedy's rule, and velocities of joints A, B, C and D using the identified instantaneous centres. Draw all instantaneous centres and velocities to scale on a separate diagram of the mechanism.
5. Analytic solutions for positions, velocities and accelerations by vector loop equations and complex number notation, and present the results in a tabular form. Compare these results with those obtained using the graphical approach. You should have good correlation.
6. All dynamic forces at the joints for the given position of the mechanism using the analytical matrix method. Assume, for link 2: $m_2$=1 kg, CG at OA/2, $I_2$=0.002 kgm2; for link 3: $m_3$=2.5 kg, CG at β/2 and AB/2, $I_3$=0.008 kgm2; for link 4: $m_4$=1.5 kg, CG at EB/2, $I_4$=0.005 kgm2; for link 5: $m_5$=1.8 kg, CG at CD/2, $I_5$=0.006 kgm2; for slider 6: $m_6$=0.9 kg and CG at D.
7. Develop an equivalent computer model capable of simulating the motion of the given mechanism using software *Working Model* and compute and plot the paths, velocities and accelerations over one revolution of the crank at the given angular velocity $\omega$. Also, find the pin forces, slider side loads and driving torque over one revolution. How do the results compare with those obtained using the analytical and graphical approach?
8. Determine the shaking force and moment and work out analytically an optimal strategy for balancing of the given mechanism. Discuss the results.

The report should be prepared under the same sections as given above and should include all working including graphical, analytical and computer simulations.

Note:

Include disc with computer model and results together with your report. Guide for modelling and simulation of the four-bar linkage using MSC Adams is attached.

Guidelines for choosing the remaining dimensions of the mechanism:

$$\overline{OE} = \overline{CD} = 4\overline{OA}$$

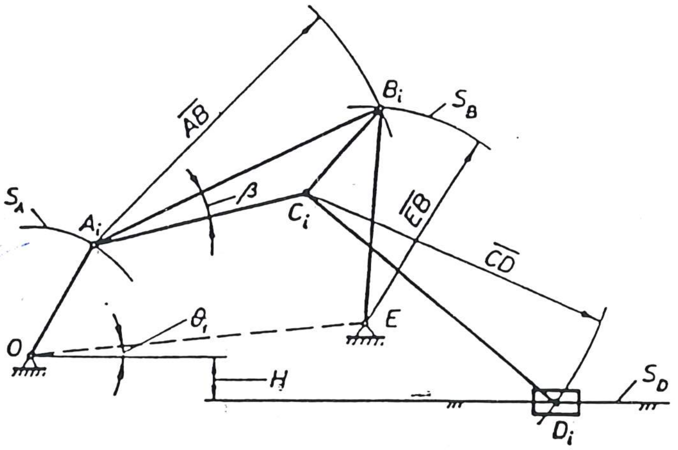
$$\overline{OA} + \overline{AB} < \overline{OE} + \overline{BE}$$

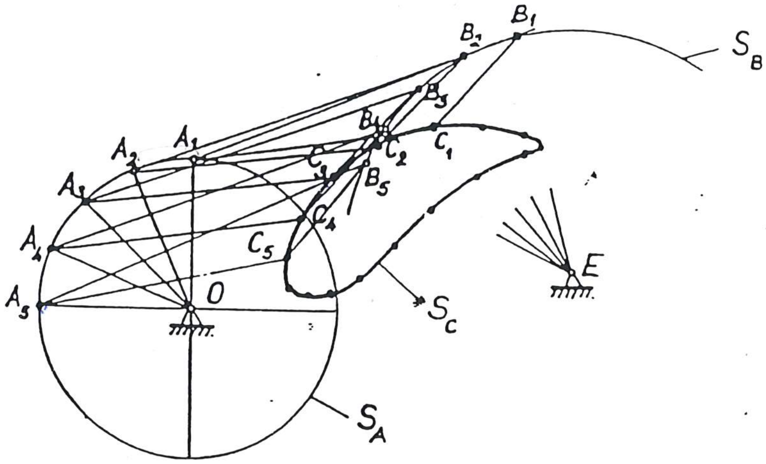
$$\overline{OA} + \overline{OE} < \overline{AB} + \overline{EB}$$

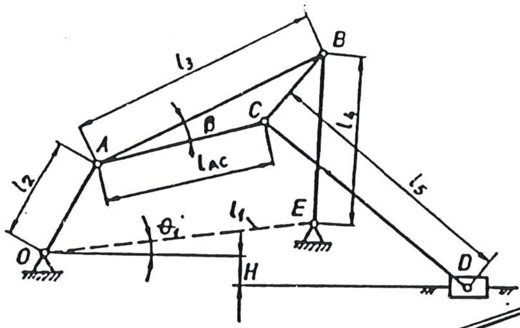
$$\overline{AC} = \frac{3}{5}\overline{AB}$$

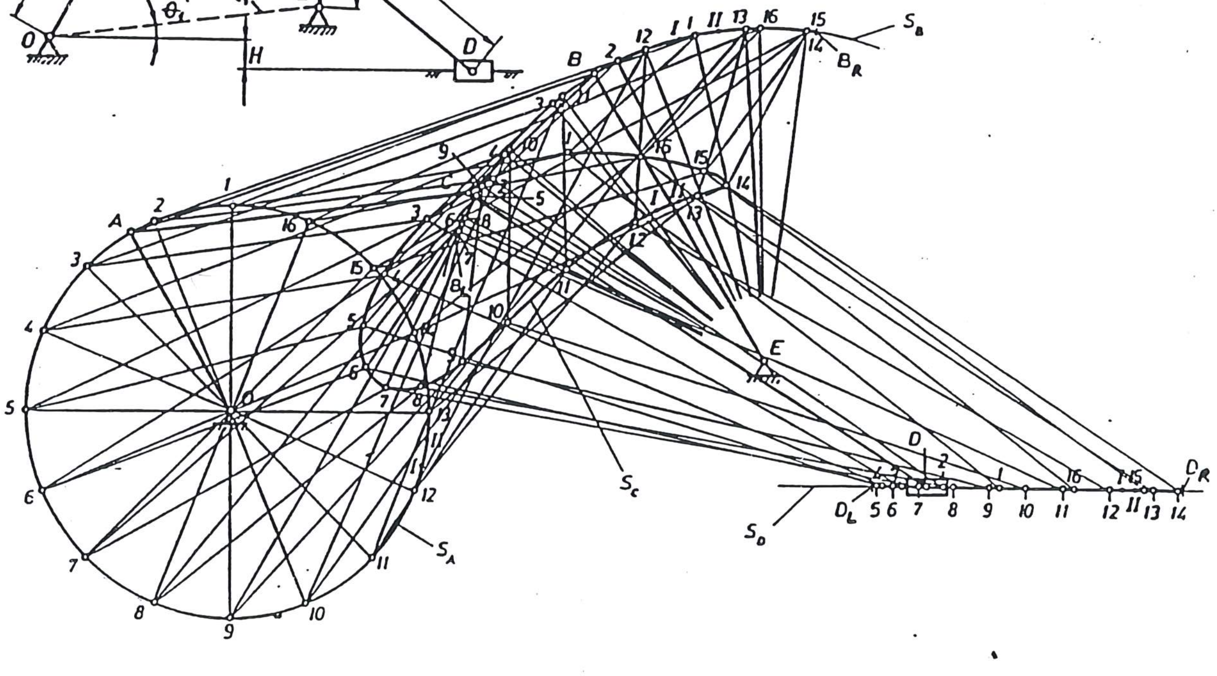
$$H = \frac{\overline{OA}}{4}$$

# GRAPHICAL POSITION ANALYSIS

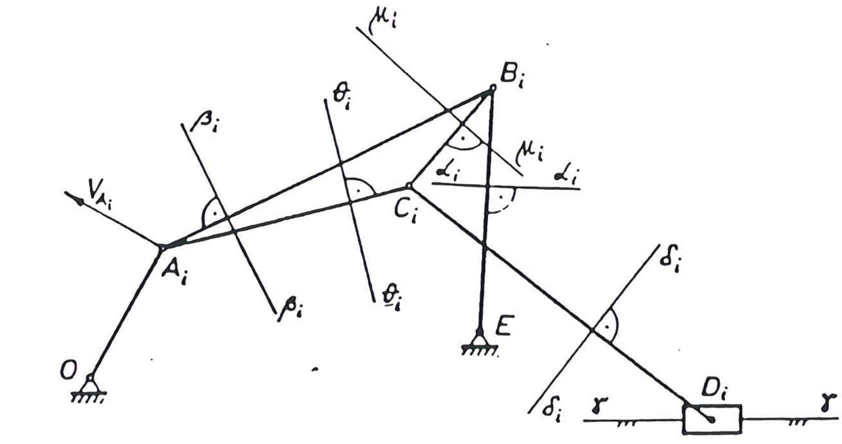


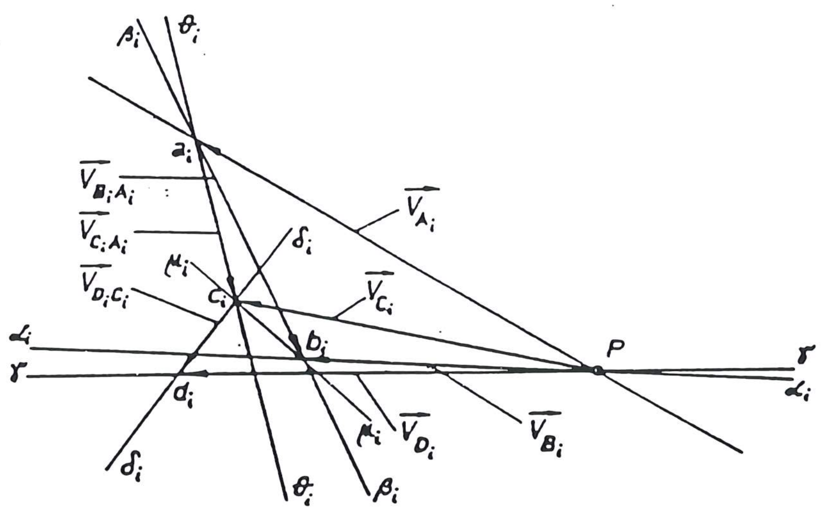






# GRAPHICAL VELOCITY ANALYSIS




$$\frac{\overline{A_iB_i}}{\overline{A_iC_i}} = \frac{\overline{a_ib_i}}{\overline{a_ic_i}}$$

$$\frac{\overline{A_iB_i}}{\overline{B_iC_i}} = \frac{\overline{a_ib_i}}{\overline{b_ic_i}}$$

$$V_A = \overline{OA}\cdot\omega_2 = \overline{OA}\cdot n_A\cdot\frac{\pi}{30} \quad |ms^{-1}|$$

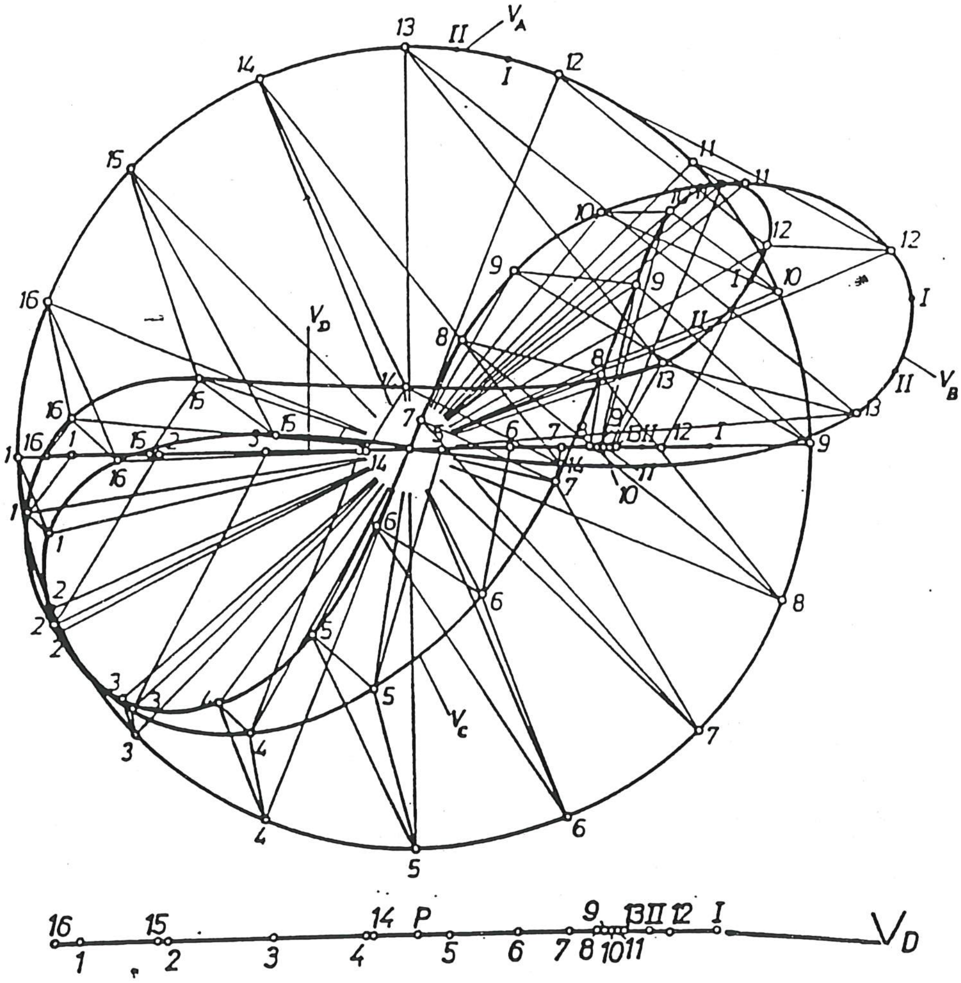
$$\vec{V}_{B_i} = \vec{V}_{A_i} + \vec{V}_{B_iA_i}$$

$$\vec{V}_{D_i} = \vec{V}_{C_i} + \vec{V}_{D_iC_i}$$

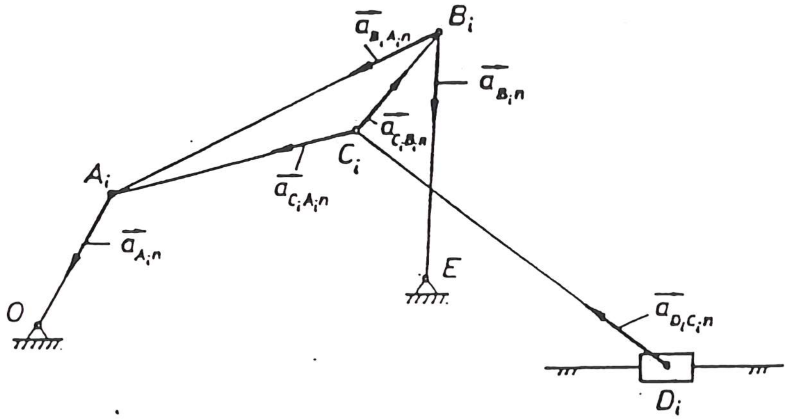
$$\vec{V}_{C_i} = \vec{V}_{A_i} + \vec{V}_{C_iA_i}$$

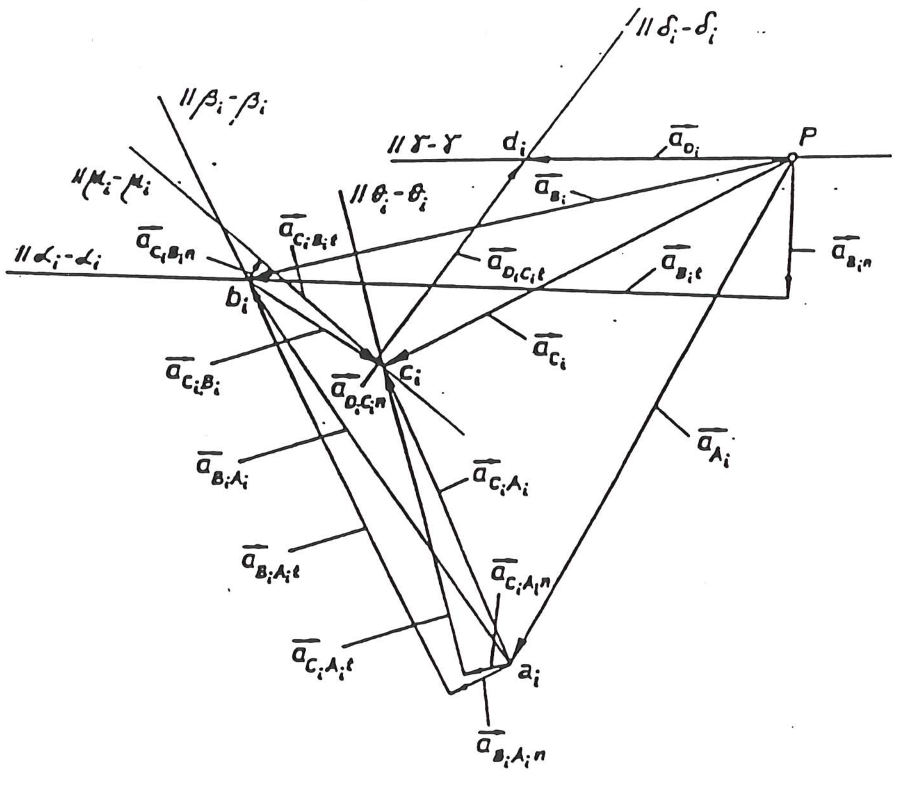
$$\vec{V}_{C_i} = \vec{V}_{B_i} + \vec{V}_{C_iB_i}$$

# VELOCITY DIAGRAM OF THE MECHANISM

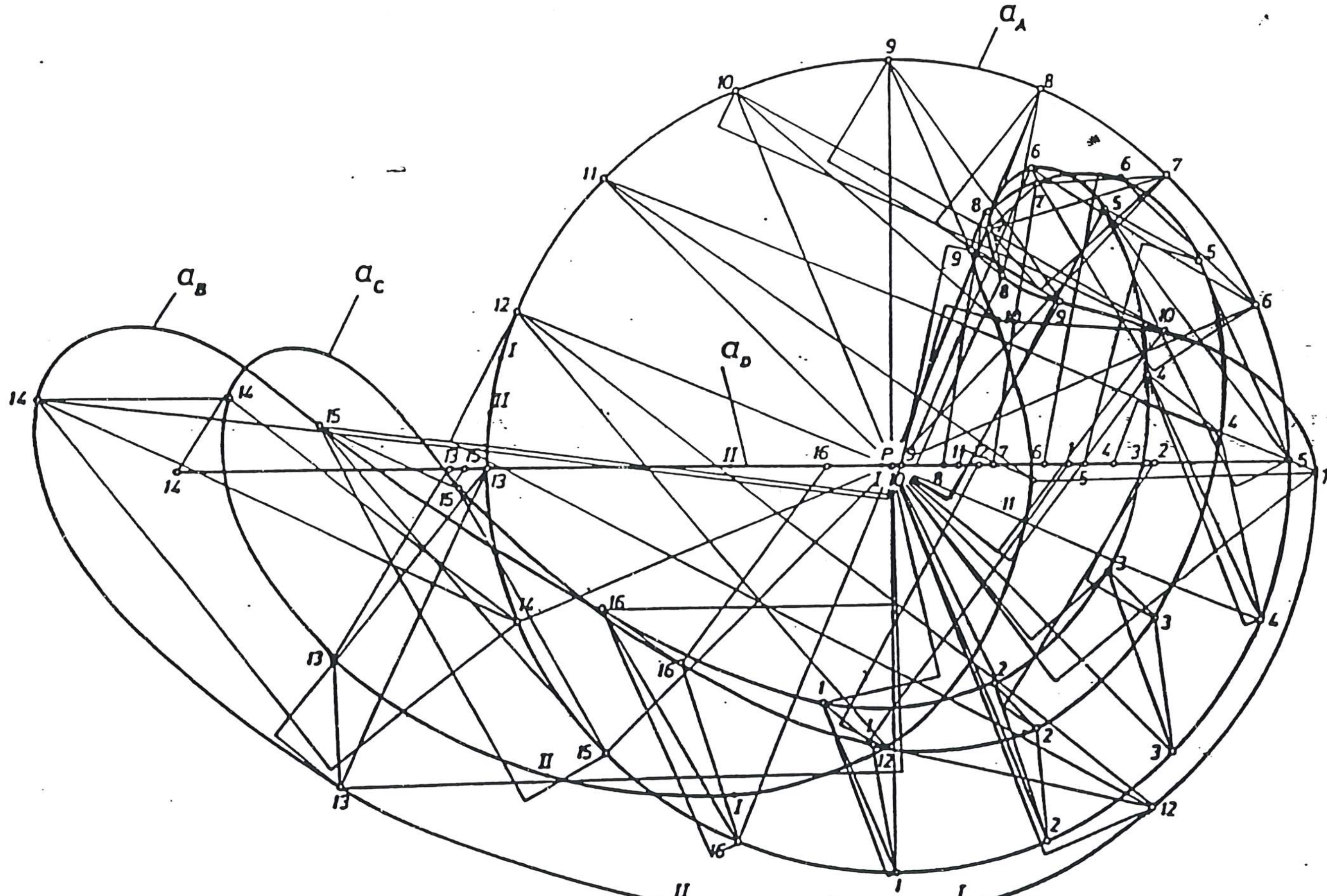


## GRAPHICAL ACCELERATION ANALYSIS

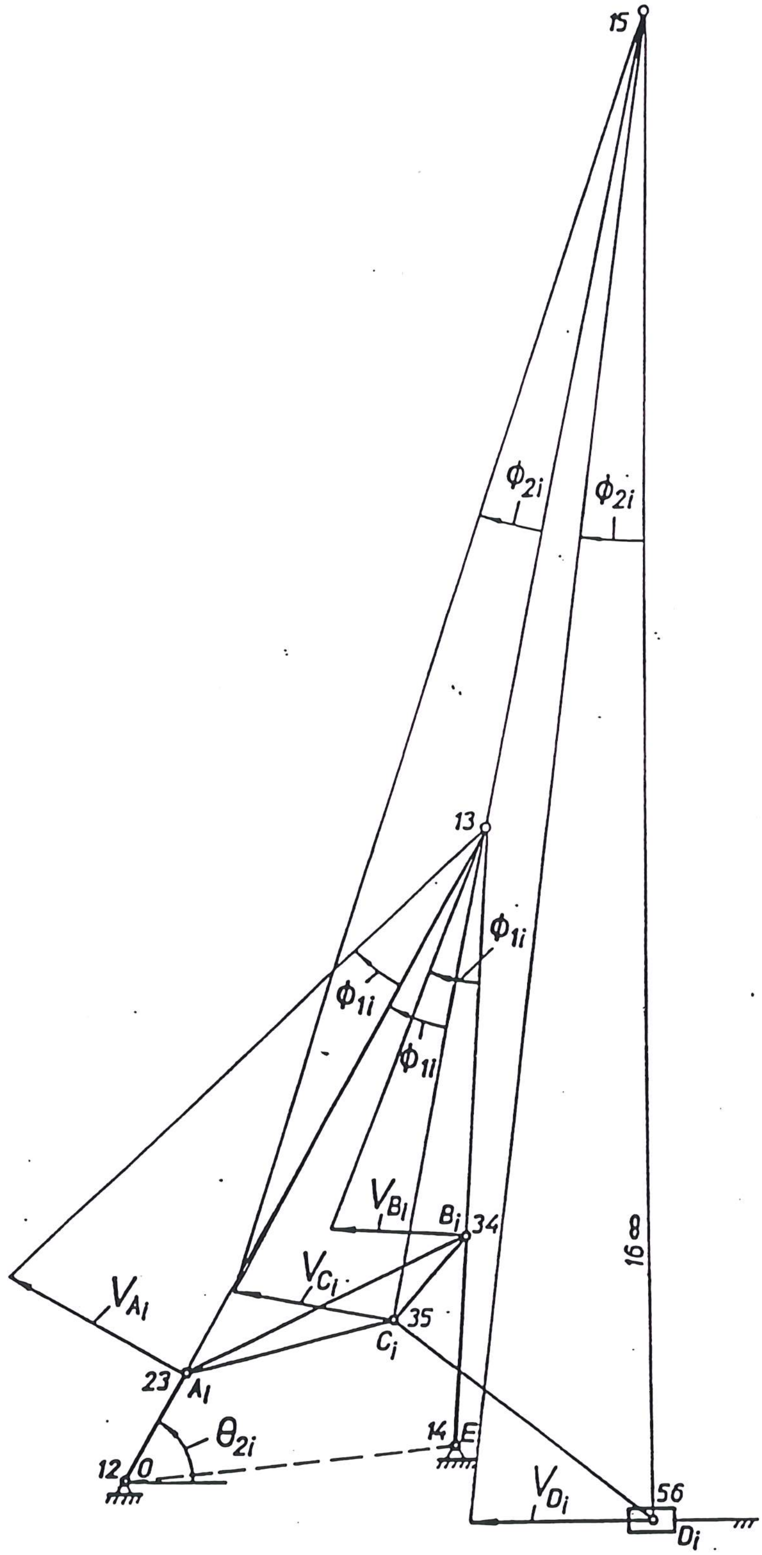


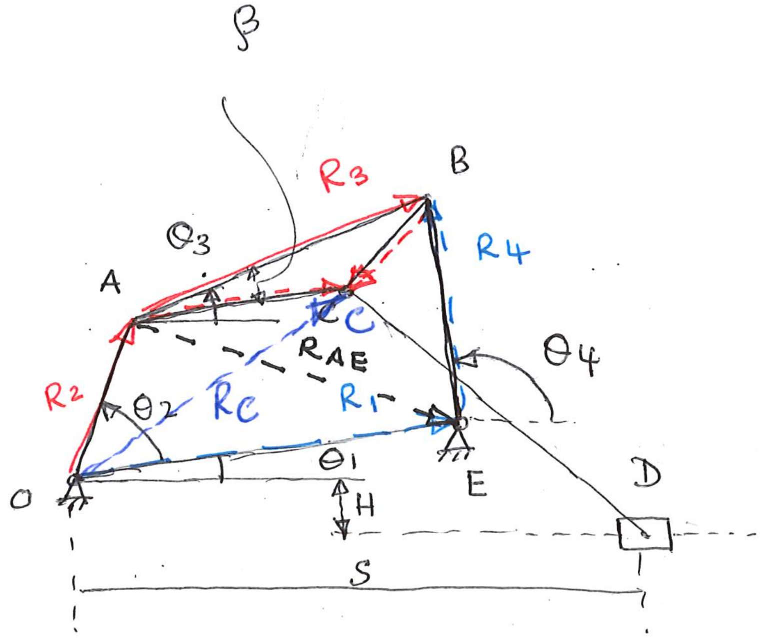



# ACCELERATION DIAGRAM OF THE MECHANISM



# VELOCITY ANALYSIS USING INSTANT CENTRES






$$R_2 + R_3 = R_1 + R_4 \quad \cdots \quad (1)$$

$$R_3 = R_{AE} + R_4 \quad \cdots \quad (2)$$

$$R_{AC} = R_3 + R_{BC} \quad \cdots \quad (3)$$

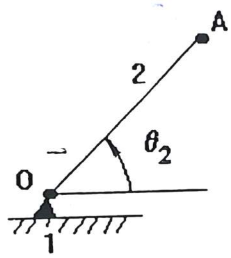
$$R_C = R_2 + R_{AC} \quad \cdots \quad (4)$$

or $R_C = R_A + R_{AC}$

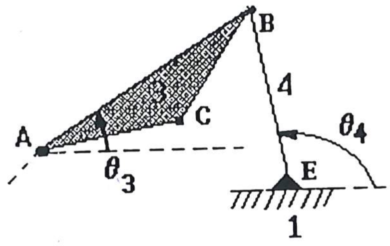
The aim of the analytical solution is to determine the position, velocity and acceleration function for each joint (A, B, C and D) as well as the angular velocity and angular acceleration functions for the coupler, the slider and each linkage. The resulting functions are to be evaluated at a number of points and the output compared with that from the graphical method.

When highly accurate results are necessary or the kinematic analysis has to be repeated for a large number of configurations, the analytical approach is best. The Assur-Artobolevsky method is based on the principle of analysing the mechanism as a set of kinematic pairs. This is essential for our example because it includes a coupler and a slider. This is a complex mechanism in terms of its motion.

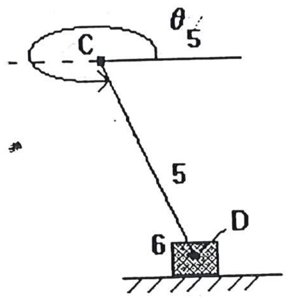
In order to solve the complex six-bar linkage, we have to decompose the mechanism into three kinematic pairs:




*Part 1* Crank




*Part 2* Coupler & Rocker




*Part 3* Slider

*Part 1* Crank (point A)

Since the underlying four-bar linkage is Grashof, the crank $OA$ will complete a full revolution. Hence, the position of the endpoint $A$ is given by

$$x_A = l_2 \cos\theta_2$$
$$y_A = l_2 \sin\theta_2$$
$$l_2 = \overline{OA} \Rightarrow l_2 = \sqrt{l_2^2 \cos^2\theta_2 + l_2^2 \sin^2\theta_2}$$
$$l_2 = \sqrt{l_2^2 (\cos^2\theta_2 + \sin^2\theta_2)} = l_2(1) = l_2$$

It follows that the components of velocity are

$$V_{Ax} = \dot{x}_A = -l_2\dot{\theta}_2 \sin\theta_2 \quad \text{and} \quad \dot{y}_A = l_2\dot{\theta}_2 \cos\theta_2 = V_{Ay}$$

$$V_A = \sqrt{V_{Ax}^2 + V_{Ay}^2}$$

$$\Rightarrow |v_A| = \sqrt{\dot{x}_A^2 + \dot{y}_A^2} = l_2|\dot{\theta}_2|$$

$$\delta_{v_A} = \theta_2 + \frac{\pi}{2} \quad \text{(velocity angle)}$$

$$\dot{\theta}_2 = \omega_2 \quad \text{(angular velocity)}$$

Similarly, the acceleration at the point $A$ can be found as follows:

$$\ddot{x}_A = -l_2\ddot{\theta}_2 \sin\theta_2 - l_2\dot{\theta}_2^2 \cos\theta_2 \quad \text{and}$$
$$\ddot{y}_A = l_2\ddot{\theta}_2 \cos\theta_2 - l_2\dot{\theta}_2^2 \sin\theta_2$$
$$\Rightarrow |a_A| = \sqrt{\ddot{x}_A^2 + \ddot{y}_A^2}$$

$$\delta_{v_A} = \arctan\left(\frac{\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2}{-(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2)}\right) \quad \text{(acceleration angle)}$$

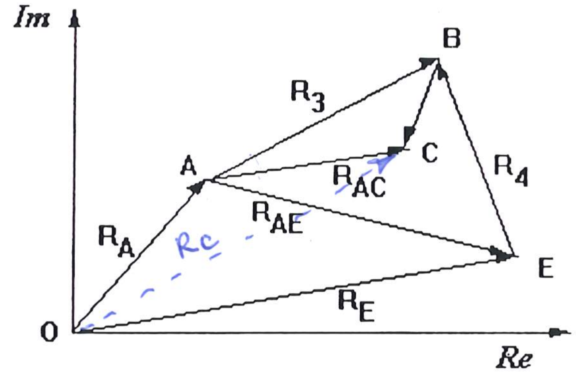
$$\ddot{\theta}_2 = \alpha_2 \quad \text{(angular acceleration)}$$

*Part 2* Coupler & Rocker (points B and C)

Because the coupler undergoes complex motion, we will represent the links as position vectors. The planar vectors for position, velocity and acceleration can then be represented in complex number form using Euler's identity

$$e^{i\theta} = \cos\theta + i\sin\theta .$$

Thus, the position vectors in the four-bar linkage consisting of crank, rocker and coupler can be represented graphically as




In order to obtain expressions for the angles $\theta_3$ and $\theta_4$, we use the above diagram to write

$$R_A + R_3 = R_E + R_4 \quad and \quad R_3 = R_{AE} + R_4$$
$$\Rightarrow R_3 = R_E + R_4 - R_A = (R_E - R_A) + R_4$$
$$Now \;\; R_{AE} = R_E - R_A = l_1 e^{i\theta_1} - l_2 e^{i\theta_2}$$
$$= l_1\cos\theta_1 - l_2\cos\theta_2 + i(l_1\sin\theta_1 - l_2\sin\theta_2)$$
$$= (\text{Real component})_{AE} + i(\text{Imaginary component})_{AE}$$

For convenience we will now define the real and imaginary components as follows:

$$\text{Re} = \text{Real component} = l_1\cos\theta_1 - l_2\cos\theta_2$$
$$\text{Im} = \text{Imaginary component} = l_1\sin\theta_1 - l_2\sin\theta_2$$

This then leads to

$$l_3 e^{i\theta_3} = \text{Re} + i\,\text{Im} + l_4 e^{i\theta_4} \quad and$$
$$l_3 e^{-i\theta_3} = \text{Re} - i\,\text{Im} + l_4 e^{i\theta_4}$$
$$\Rightarrow l_3^2 = \text{Re}^2 + \text{Im}^2 + \text{Re}\, l_4(\cos\theta_4 + i\sin\theta_4) - i\,\text{Im}\, l_4(\cos\theta_4 + i\sin\theta_4) .$$
$$+ \text{Re}\, l_4(\cos\theta_4 - i\sin\theta_4) + i\,\text{Im}\, l_4(\cos\theta_4 + i\sin\theta_4) + l_4^2$$
$$\Rightarrow \text{Re}^2 + \text{Im}^2 + 2\,\text{Re}\, l_4\cos\theta_4 + 2\,\text{Im}\, l_4\sin\theta_4 + l_4^2 - l_3^2 = 0$$

From this we can obtain the following expressions

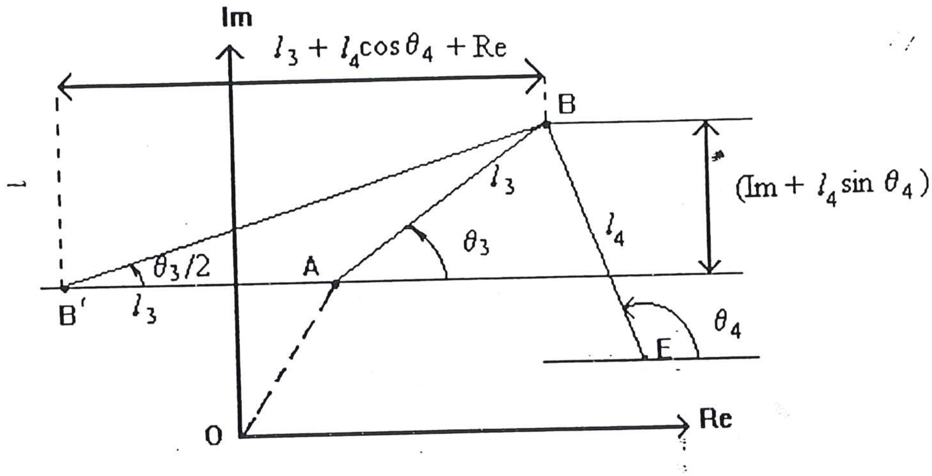
$$\sin\theta_4 = \frac{2\tan\frac{\theta_4}{2}}{1+\tan^2\frac{\theta_4}{2}} \quad \text{and} \quad \cos\theta_4 = \frac{1-\tan^2\frac{\theta_4}{2}}{1+\tan^2\frac{\theta_4}{2}} .$$

If we now let $C = \dfrac{\mathrm{Re}^2 + \mathrm{Im}^2 + l_4^2 - l_3^2}{2 l_4}$ , then we obtain the following

$$\left(\tan\frac{\theta_4}{2}\right)_{1,2} = \frac{-\mathrm{Im} \pm \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2 - C^2}}{C - \mathrm{Re}} \qquad (where \;\; 0 < \theta_4 < \pi)$$

$$\Rightarrow \theta_4 = 2\arctan\left(\frac{-\mathrm{Im} \pm \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2 - C^2}}{C - \mathrm{Re}}\right)$$

Since we want an open configuration, we will take the positive value. To obtain an expression for $\theta_3$, we require the following diagram.




From the angle $AB'B$, we obtain

$$\theta_3 = 2\arctan\left(\frac{\mathrm{Im} \pm l_4 \sin\theta_4}{l_3 + \mathrm{Re} + l_4 \cos\theta_4}\right).$$

Finally, we obtain the following expressions for $X_B$ and $X_E$ :

$$X_B = X_E + l_4 \cos\theta_4 \quad \text{and} \quad Y_B = Y_E + l_4 \sin\theta_4 .$$

Similarly, we can analyse the position of the point $C$ . Clearly,

$$R_C = R_A + R_{AC} = l_2 e^{i\theta_2} + l_{AC} e^{i(\theta_3 + \beta)} \qquad (+\beta \text{ when } C \text{ is above } AB).$$

This leads to the following expressions for $X_C$ and $Y_C$ :

$$X_C = l_2 \cos\theta_2 + l_{AC} \cos(\theta_3 + \beta)$$

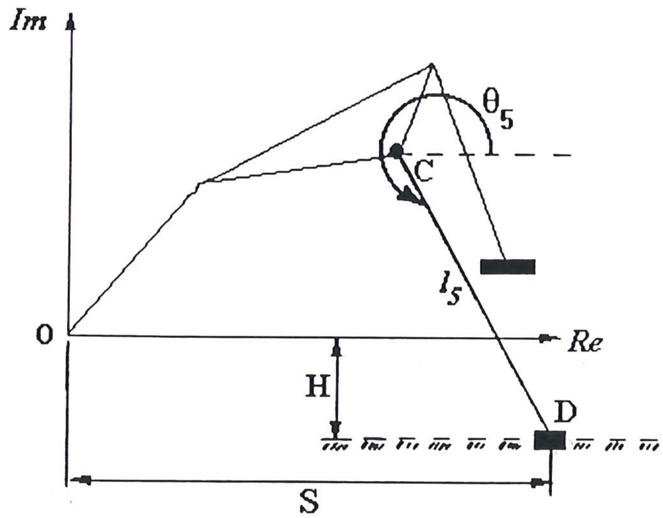
and

$$Y_C = l_2 \sin\theta_2 + l_{AC} \sin(\theta_3 + \beta)$$

Differentiation then yields expressions for the velocity and acceleration components.

*Part 3* Slider (point D)

The final part of the analysis concerns the position of the slider (point D). The following diagram required for this analysis.




From the diagram it follows that

$$X_C + iY_C + l_5(\cos\theta_5 + i\sin\theta_5) = He^{i\pi/2} + S = iH + S$$
$$\Rightarrow X_C + l_5\cos\theta_5 = S \quad \text{and} \quad Y_C + l_5\sin\theta_5 = H$$
$$\Rightarrow \sin\theta_5 = \frac{H - Y_C}{l_5}$$
$$\Rightarrow \theta_5 = \arcsin\frac{H - Y_C}{l_5}$$

Again, velocities and accelerations can be found by differentiating the appropriate functions. Finally,

$$X_D = S = X_C + l_5\cos\theta_5 \quad \text{and} \quad Y_D = -H\,.$$

$$\frac{d}{dt}(x_D) = V_{xD}$$

$$\frac{d}{dt}(Y_D) = V_{yD}$$

$$V_D = \sqrt{(V_{xD})^2 + (V_{yD})^2}$$

# *Animation in Maple*

Once you have defined the $x$ and $y$ components for each of the points $A$, $B$, $C$, $D$ and $E$, you can form expressions for each of the linkages in terms of the position vectors of each of these points. For example, the coupler is the triangle $ABC$ and so to describe the motion of the coupler we need to determine the position vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$.

Now the points on the vector $\overrightarrow{AC}$ have for their $x$ component a linear combination of the $x$ components of the points $A$ and $C$. Similarly for their $y$ components. That is, we can define the vector as follows:

$$[x_A + u * (x_C - x_A),\ y_A + u * (y_C - y_A)]$$

where $u$ takes values between 0 and 1. We will use $u$ as the animation parameter in a plot of the vector $\overrightarrow{AC}$ and use a time period equal to that of the driver link. For example,

*anim1 := animate([xA + u*(xC - xA), yA + u*(yC - yA), u = 0..1], t=0..2):*

Following the above procedure, generate each of the six animations required for the mechanism but suppress the output by using a colon at the end of the statement. Display all six graphs on the one set of axes. Before running the animation, re-size the graph window to make sure that the axes have the same scale otherwise there may appear to be distortion of the coupler.

# SYMBOLIC PROCEDURE FOR POSITION, VELOCITY AND ACCELERATION ANALYSIS

```
> Group_All:=proc(x1,x2,x3,x4,x5,x6,x7,x8)
> with(linalg): with(plots):
> L1:=x1:L2:=x2:L3:=x3:L4:=x4:L5:=x5:LAC:=x6:H:=x7:beta:=x8:
> n2:=553:theta1:=Pi/10:
> omega2:=2*Pi*n2/60:
> theta2(t):='theta2(t)':
> de1:=diff(theta2(t),t)=omega2:
> desol:=dsolve(de1,theta2(t)):
> subs(t=0, theta2(0)=60*Pi/180, desol):
> theta2(t):=553*Pi/30*t+Pi/3:
> XA:=L2*cos(theta2(t)):
> YA:=L2*sin(theta2(t)):
> DXA:=diff(XA,t$1):
> DYA:=diff(YA,t$1):
> DDXA:=diff(XA,t$2):
> DDYA:=diff(YA,t$2):
> RAEreal:=L1*cos(theta1)-L2*cos(theta2(t)):
> RAEimag:=L1*sin(theta1)-L2*sin(theta2(t)):
> Constant:=((RAEreal)^2+(RAEimag)^2+L4^2-L3^2)/(2*L4):
> theta4(t):=2*arctan((-RAEimag-sqrt(RAEreal^2+RAEimag^2-Constant^2))/(Constant-RAEreal)):
> theta3(t):=2*arctan((RAEimag+L4*sin(theta4(t)))/(L3+RAEreal+L4*cos(theta4(t)))):
> XB:=L1*cos(theta1)+L4*cos(theta4(t)):
> YB:=L1*sin(theta1)+L4*sin(theta4(t)):
> DXB:=diff(XB,t$1):
> DYB:=diff(YB,t$1):
> DDXB:=diff(XB,t$2):
> DDYB:=diff(YB,t$2):
> XC:=L2*cos(theta2(t))+LAC*cos(theta3(t)+beta):
> YC:=L2*sin(theta2(t))+LAC*sin(theta3(t)+beta):
> DXC:=diff(XC,t$1):
> DYC:=diff(YC,t$1):
> DDXC:=diff(XC,t$2):
> DDYC:=diff(YC,t$2):
> theta5(t):=arcsin((H-YC)/L5):
> XD:=XC+L5*cos(theta5(t)):
> YD:=YC+L5*sin(theta5(t)):
> DXD:=diff(XD,t$1):
> DYD:=diff(YD,t$1):
> DDXD:=diff(XD,t$2):
> DDYD:=diff(YD,t$2):
> XE:=L1*cos(theta1):
> YE:=L1*sin(theta1):
> end:
```

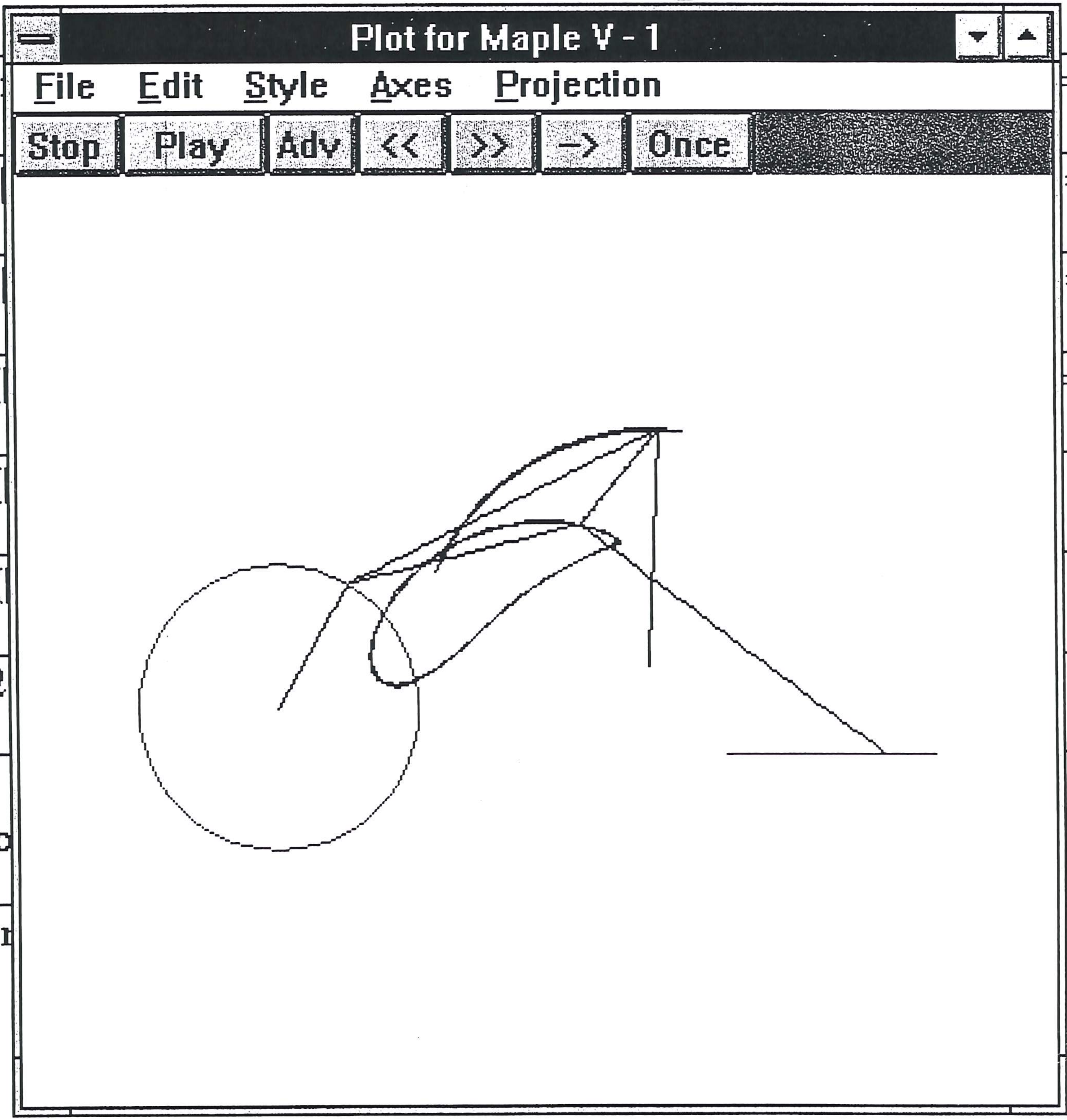
# SYMBOLIC PROCEDURE FOR ANIMATION

```
> Position_Animation:=proc()
> LinkAC:=animate([XA+d*(XC-XA),YA+d*(YC-YA),d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> LinkBC:=animate([XB+d*(XC-XB),YB+d*(YC-YB),d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> LinkAB:=animate([XA+d*(XB-XA),YA+d*(YB-YA),d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> LinkBE:=animate([XB+d*(XE-XB),YB+d*(YE-YB),d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> LinkCD:=animate([XC+d*(XD-XC),YC+d*(YD-YC),d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> LinkOA:=animate([(XA)*d,(YA)*d,d=0..1],
> t=0..0.11,axes=none,color=black,frames=25,numpoints=2):
> PathA:=animate([XA,YA,t=0..0.12],d=0..1,
> color=blue,axes=NONE,frames=25):
> PathB:=animate([XB,YB,t=0..0.12],d=0..1,
> color=violet,axes=NONE,frames=25):
> PathC:=animate([XC,YC,t=0..0.12],d=0..1,
> color=red,axes=NONE,frames=25):
> PathD:=animate([XD,YD,t=0..0.12],d=0..1,
> color=brown,axes=NONE,frames=25,title='Position_Animation'):
> display({LinkOA,LinkAB,LinkAC,LinkBC,LinkCD,LinkBE,PathA,PathB,PathC,PathD});
> end:
```

Maple V for Windows - LINK2.MS

File Edit Format Options Help

Below is the animation of the six-bar linkage mechanism.

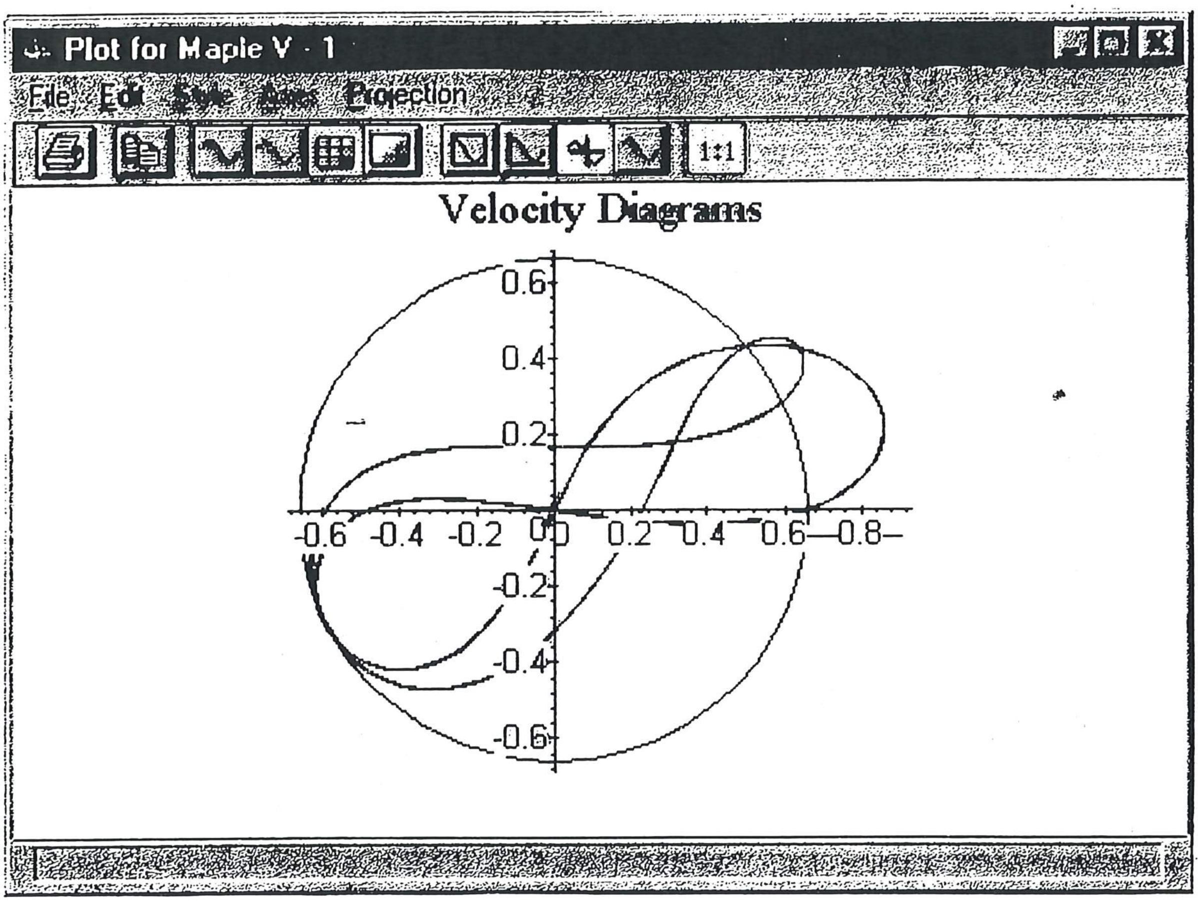



```
> a1 :=animate([  = 20):
> a2 := animate(  = 20):
> a3 := animate(  = 20):
> a4 := animate(  = 20):
> a5 := animate(  = 20):
> a6 := animate(
> display({a1, a2
```

Tables of co

```
> pos_table := pr
>
>
>     printf(`\n t
```

# VELOCITY DIAGRAMS OF SIX-BAR LINKAGE MECHANISM



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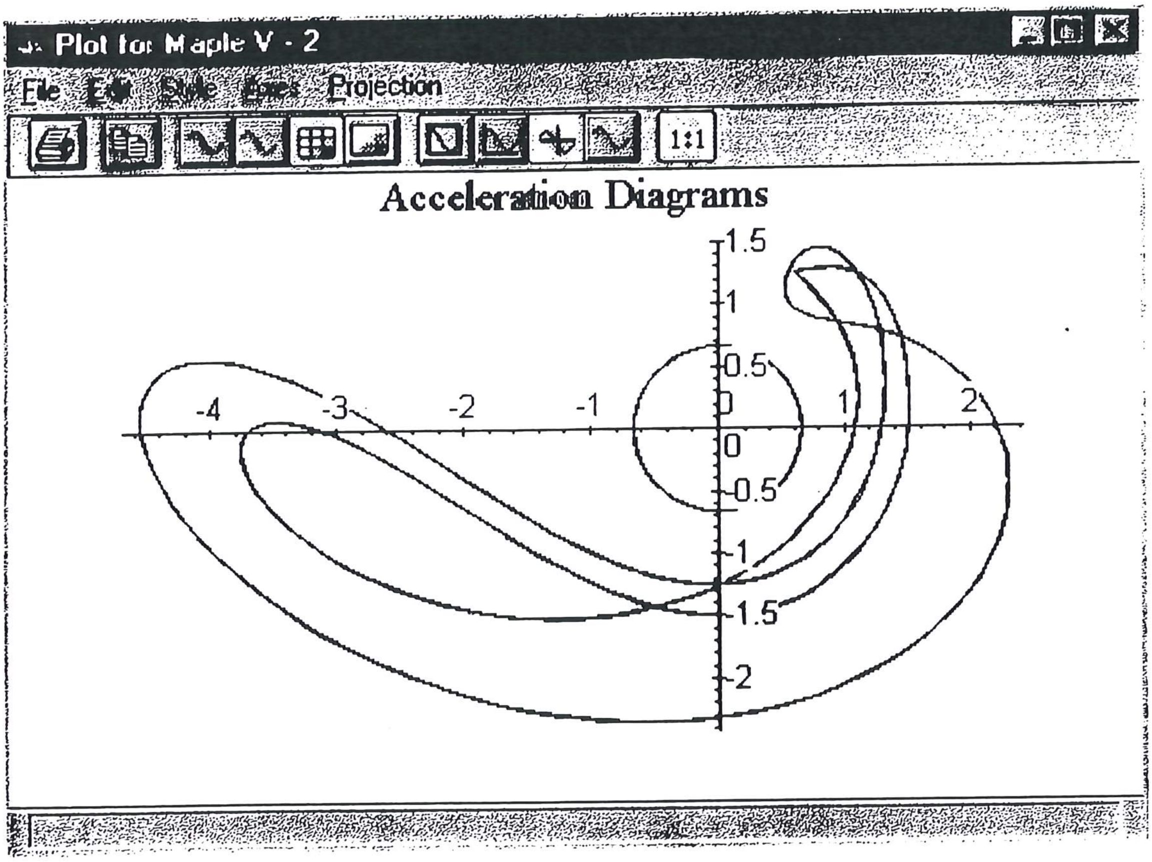
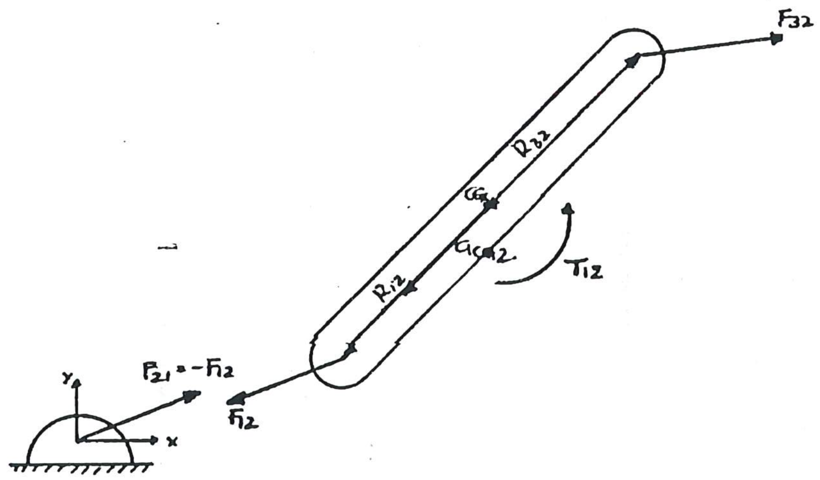


# Dynamic Force Analysis

**Analysing link 2:**




**Link 2 - free-body diagram**

Summing the forces yields,

$$\sum F = F_{12} + F_{32} = m g_{G2}$$

Unknowns:

$F_{12X}, F_{12Y}, F_{32X}, F_{32Y}, T_{12}$

Summing the Torque yields,

$$\sum T = T_{12} + (R_{12} \times F_{12}) + (R_{32} \times F_{32}) = I_{G2}\alpha$$

Representing Force as X-components;

$$F_{12X} + F_{32X} = m g_{G2X}$$

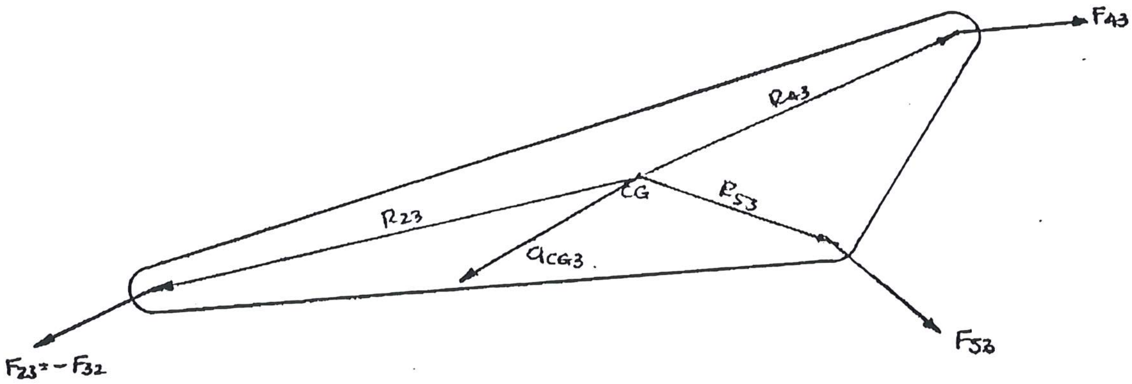
Representing Force as Y-components;

$$F_{12Y} + F_{32Y} = m g_{G2Y}$$

Representing Torque as:

$$T_{12} + (R_{12X} \times F_{12Y} - R_{12Y} \times F_{12X}) + (R_{32X} \times F_{32Y} - R_{32Y} \times F_{32X}) = I_{G2}\alpha$$

Analysing link 3:




Link 3 - free-body diagram

Summing the forces yields,

$$\sum F = F_{43} - F_{32} + F_{53} = mg_{G3}$$

Unknowns:

$F_{32X}, F_{32Y}, F_{43X}, F_{43Y}, F_{53X}, F_{53Y}$

Summing the Torque yields,

$$\sum T = (R_{23} \times F_{23}) + (R_{53} \times F_{53}) + (R_{43} \times F_{43}) = I_{G3}\alpha$$

Representing Force as X-components;

$$F_{43X} - F_{32X} + F_{53X} = mg_{G3X}$$

Representing Force as Y-components;

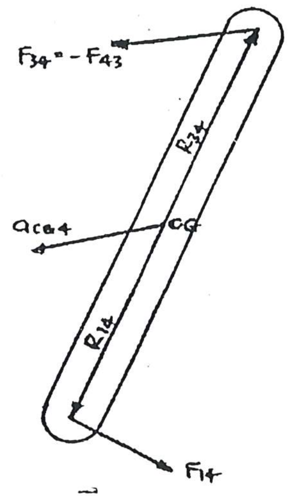
$$F_{43Y} - F_{32Y} + F_{53Y} = mg_{G3Y}$$

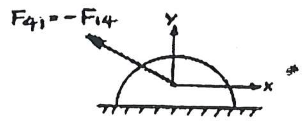
Representing Torque as: ($F_{23} = -F_{32}$)

$$(R_{23X} \times F_{23Y} - R_{23Y} \times F_{23X}) + (R_{53X} \times F_{53Y} - R_{53Y} \times F_{53X}) + (R_{43X} \times F_{43Y} - R_{43Y} \times F_{43X}) = I_{G3}\alpha$$

$$-(R_{23X} \times F_{32Y} - R_{23Y} \times F_{32X}) + (R_{53X} \times F_{53Y} - R_{53Y} \times F_{53X}) + (R_{43X} \times F_{43Y} - R_{43Y} \times F_{43X}) = I_{G3}\alpha$$

**Analysing link 4:**




**Link 4 - free-body diagram**

Summing the forces yields,

$$\sum F = F_{14} - F_{43} = mg_{G4}$$

Unknowns:

$F_{14X}, F_{14Y}, F_{43X}, F_{43Y}$

Summing the Torque yields,

$$\sum T = (R_{34} \times F_{34}) + (R_{14} \times F_{14}) = I_{G4}\alpha$$

Representing Force as X-components;

$$F_{14X} - F_{43X} = mg_{G4X}$$

Representing Force as Y-components;

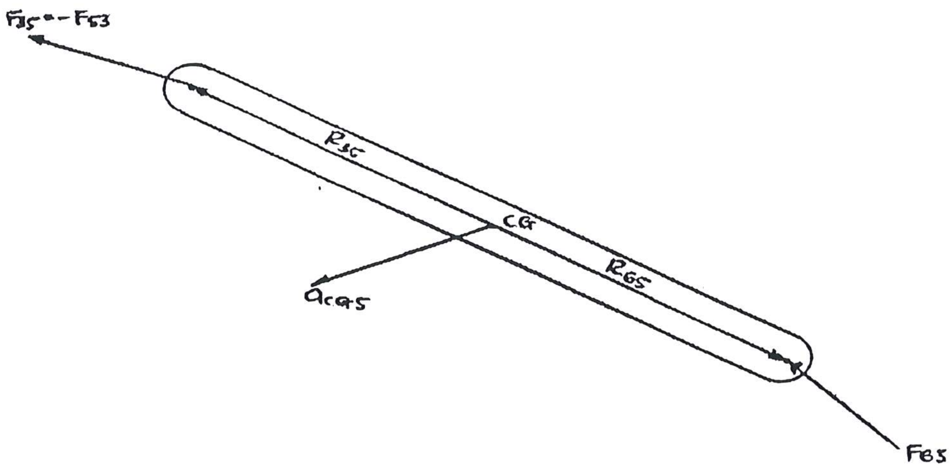
$$F_{14Y} - F_{43Y} = mg_{G4Y}$$

Representing Torque as: ($F_{34} = -F_{43}$)

$$(R_{34X} \times F_{34Y} - R_{34Y} \times F_{34X}) + (R_{14X} \times F_{14Y} - R_{14Y} \times F_{14X}) = I_{G4}\alpha$$

$$-(R_{23X} \times F_{43Y} - R_{23Y} \times F_{43X}) + (R_{14X} \times F_{14Y} - R_{14Y} \times F_{14X}) = I_{G4}\alpha$$

**Analysing link 5:**




**Link 5 - free-body diagram**

Summing the forces yields,

$$\sum F = F_{65} - F_{53} = mg_{G5}$$

Unknowns:

$F_{65X}, F_{65Y}, F_{53X}, F_{53Y}$

Summing the Torque yields,

$$\sum T = (R_{35} \times F_{35}) + (R_{65} \times F_{65}) = I_{G5}\alpha$$

Representing Force as X-components;

$$F_{65X} - F_{53X} = mg_{G5X}$$

Representing Force as Y-components;

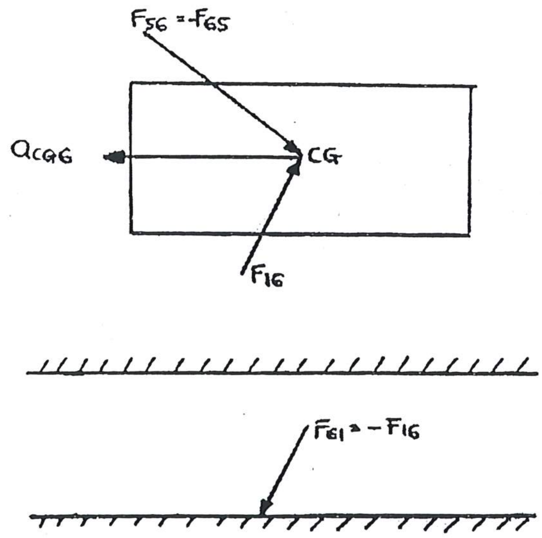
$$F_{65Y} - F_{53Y} = mg_{G5Y}$$

Representing Torque as: ($F_{35} = -F_{53}$)

$$(R_{35X} \times F_{35Y} - R_{35Y} \times F_{35X}) + (R_{65X} \times F_{65Y} - R_{65Y} \times F_{65X}) = I_{G5}\alpha$$

$$-(R_{35X} \times F_{53Y} - R_{35Y} \times F_{53X}) + (R_{65X} \times F_{65Y} - R_{65Y} \times F_{65X}) = I_{G5}\alpha$$

**Analysing link 6:**




**Link 6 - free-body diagram**

Summing the forces yields,

$$\sum F = F_{16} - F_{65} = mg_{G6}$$

Unknowns:

$F_{16X}, F_{65X}, F_{65Y}$

Summing the Torque yields,

$$\sum T = 0 \text{ (all forces through centre of mass)}$$

Representing Force as X-components;

$$F_{16X} - F_{65X} = mg_{G6X}$$

Knowing: $F_{\mu} = \pm\mu N$

$\therefore F_{16X} = 0.2F_{16Y}$

$$0.2F_{16Y} - F_{65X} = mg_{G6X}$$

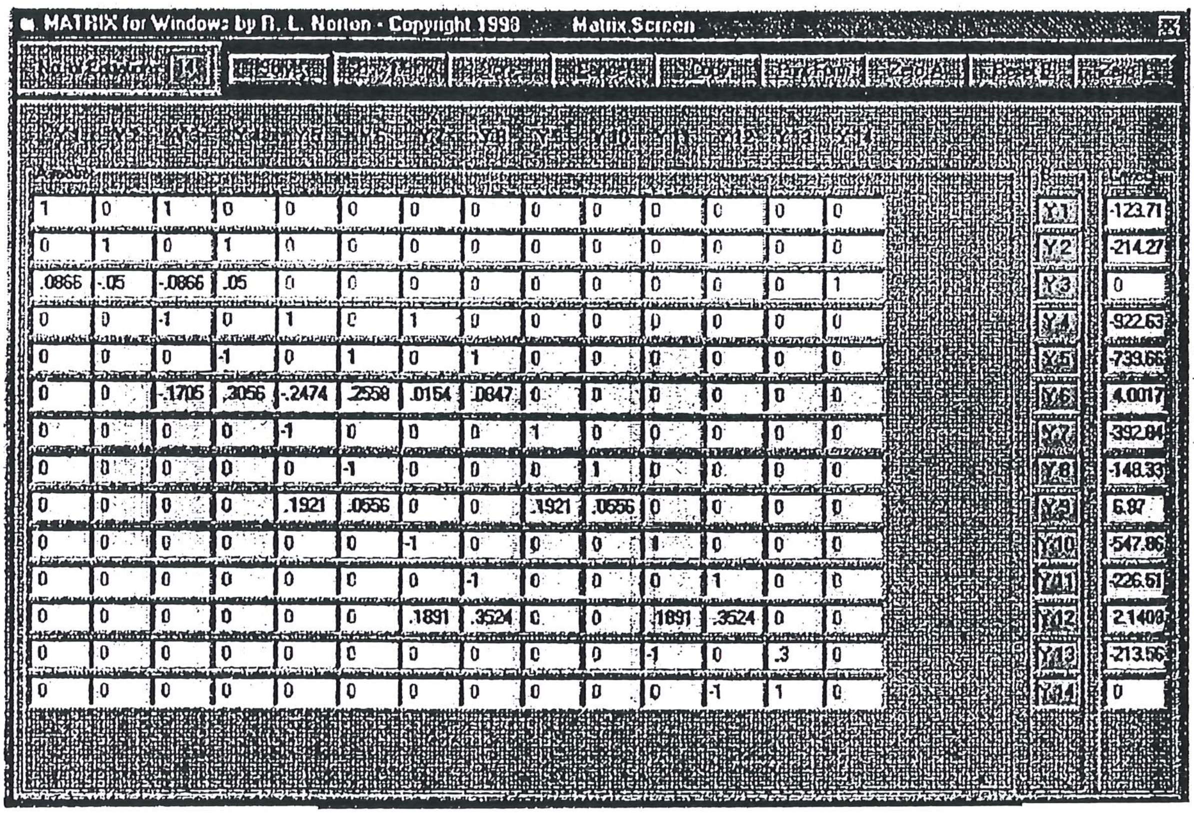
Representing Force as Y-components;

$$F_{16Y} - F_{65Y} = 0$$

## Obtaining a complete Matrix

Once the accelerations about the centre of gravity for each link have been obtained, the masses of each link, together with the angular acceleration, moment of inertia and the position vector values can be inserted into Matrix

**NOTE: The coefficient of friction for the slider has been taken to be $\mu = 0.3$.**




| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Y1 | -123.71 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Y2 | -214.27 |
| .0866 | -.05 | -.0866 | .05 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | Y3 | 0 |
| 0 | 0 | -1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Y4 | -922.63 |
| 0 | 0 | 0 | -1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | Y5 | -739.66 |
| 0 | 0 | -.1705 | .3056 | -.2474 | .2558 | .0154 | .0847 | 0 | 0 | 0 | 0 | 0 | 0 | Y6 | 4.0017 |
| 0 | 0 | 0 | 0 | -1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | Y7 | 392.84 |
| 0 | 0 | 0 | 0 | 0 | -1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | Y8 | -148.33 |
| 0 | 0 | 0 | 0 | .1921 | .0656 | 0 | 0 | .1921 | .0656 | 0 | 0 | 0 | 0 | Y9 | 6.97 |
| 0 | 0 | 0 | 0 | 0 | 0 | -1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | Y10 | 547.86 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 | 0 | 0 | 0 | 1 | 0 | 0 | Y11 | -226.51 |
| 0 | 0 | 0 | 0 | 0 | 0 | .1891 | .3524 | 0 | 0 | .1891 | .3524 | 0 | 0 | Y12 | 2.1400 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 | 0 | .3 | 0 | Y13 | -213.56 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 | 1 | 0 | Y14 | 0 |

Use program MATRIX to solve the matrix for unknown dynamic forces and driving torque

# Shaking Force & Moment

## Shaking forces

The net effect of all the dynamic forces experienced by the ground of the mechanism, translate to vibrations in the structure which supports the mechanism. The sum of all the forces acting on the ground plane is referred to as the **Shaking Force**, which for this particular case is represented as,

$$F_s = F_{21} + F_{41} + F_6 \qquad \text{given,} \quad F_{21} = -F_{12} \quad F_{41} = -F_{14,} \quad F_{61} = -F_{16}$$

where

$F_{21X} = 1990.37$ N
$F_{21y} = 999.27$ N
$F_{41X} = -4.139$ N
$F_{41Y} = -566.15$ N
$F_{61X} = 76.16$ N
$F_{61Y} = 253.97$ N

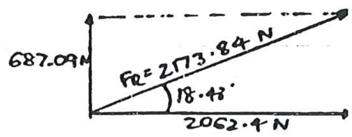
In order to summate the result force experienced by the ground, a summation was undertaken of the x and y components of the force respectively.

- Finding

$$\begin{aligned} \Sigma F_{Rx} &= F_{21X} + F_{41X} + F_{61X} \\ &= 1990.37 - 4.139 + 76.16 \\ &= 2062.4 \text{ N} \end{aligned}$$

- Finding

$$\begin{aligned} \Sigma F_{Ry} &= F_{21Y} + F_{41Y} + F_{61Y} \\ &= 999.27 - 566.15 + 253.97 \\ &= 687.09 \text{ N} \end{aligned}$$




**Resultant vector for shaking force**

Therefore,

$$F_R = \mathbf{2173.84}\ \text{N}$$

The shaking force will tend to move the structure (ground) in the transverse direction, but due to the influence of the driving link with respect to the entire mechanism shaking torque is induced to the system. The shaking torque is simply the reaction torque felt by the ground that tends to rock the structure about its driveline axis. The **Shaking Torque** is represented as,

$$T_s = T_{21} = -T_{12}$$

with $-T_{12}$ representing the negative of the source torque which is delivered by the driving link.

Therefore,

$$T_s = \mathbf{-143.491}\ \text{Nm}$$

## Shaking moments

The shaking moment of a linkage mechanism is the summation of the reaction torque $T_{21}$ and the shaking couples introduced the connections to the ground, which is represented as,

$$M_S = T_{21} + (R_1 \times F_{41}) + (R_2 \times F_{61})$$

with,

$T_{21}$ = the negative of the driving torque,
$R_1$ = position vector from ground of driver (O) to ground of link 4,
$F_{41}$ = force of link 4 onto ground.
$R_1$ = position vector from ground of driver (O) to ground of slider,
$F_{61}$ = force of slider onto ground.

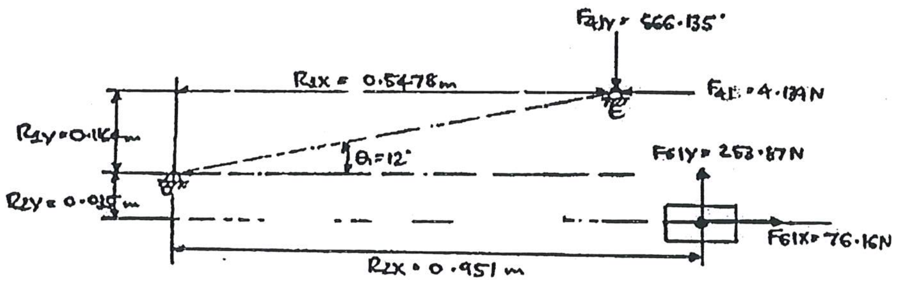



**Diagram of forces generating shaking moments**

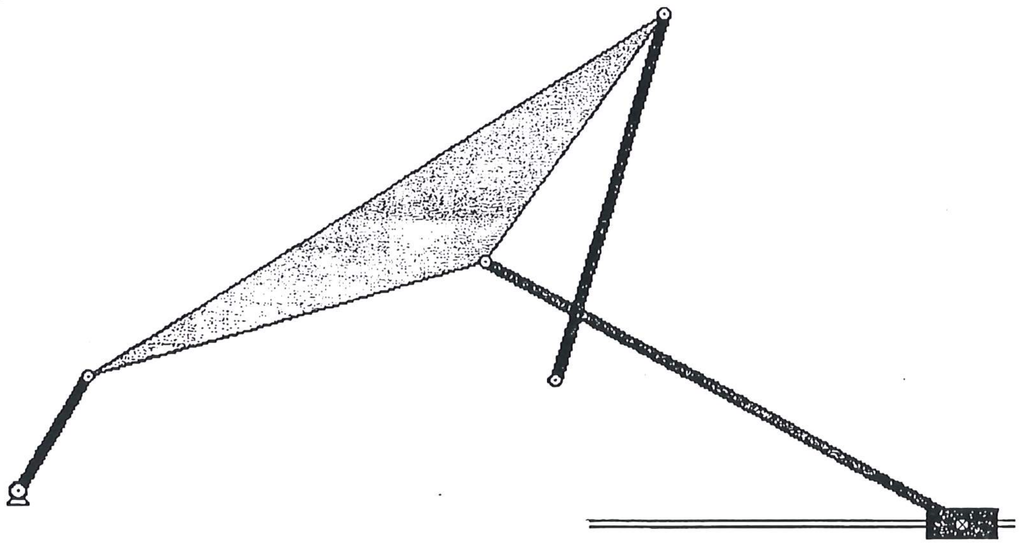
Therefore,

$$\begin{aligned} M_S &= T_{21} + (R_{1X} \times F_{41Y}) - (R_{1Y} \times F_{41X}) - (R_{2X} \times F_{61Y}) - (R_{2Y} \times F_{61X}) \\ &= 143.491 \text{ Nm} + (0.5478\text{x}566.135) - (0.1164\text{x}4.139) - (0.035\text{x}76.16) - (0.951\text{x}253.87) \\ &= \mathbf{209.04 \text{ Nm}} \end{aligned}$$

The effect of the shaking moment, is basically a rocking effect about the axis of the motor's drive shaft,

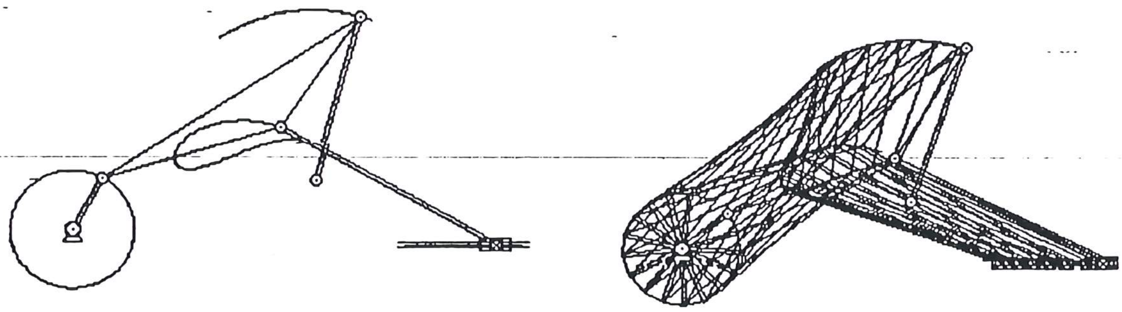
# Example of Working Model Results

The model represents the mechanism consisting of a motor, linkages, pin joints, slider and a slot for the slider to travel along. In order to achieve one revolution, the time incrementation was set for one frame of 16, using the 'Accuracy' command, and then by setting the 'Pause Control ' for 16 frames the desirable stopping location was obtained.



**Working Model**

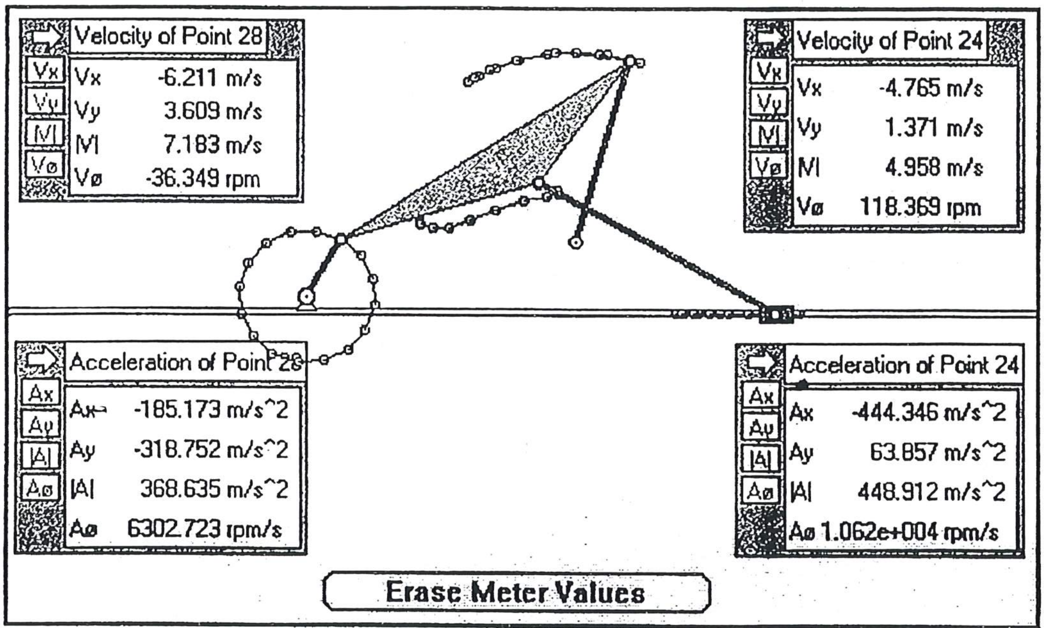
## Plot of Path



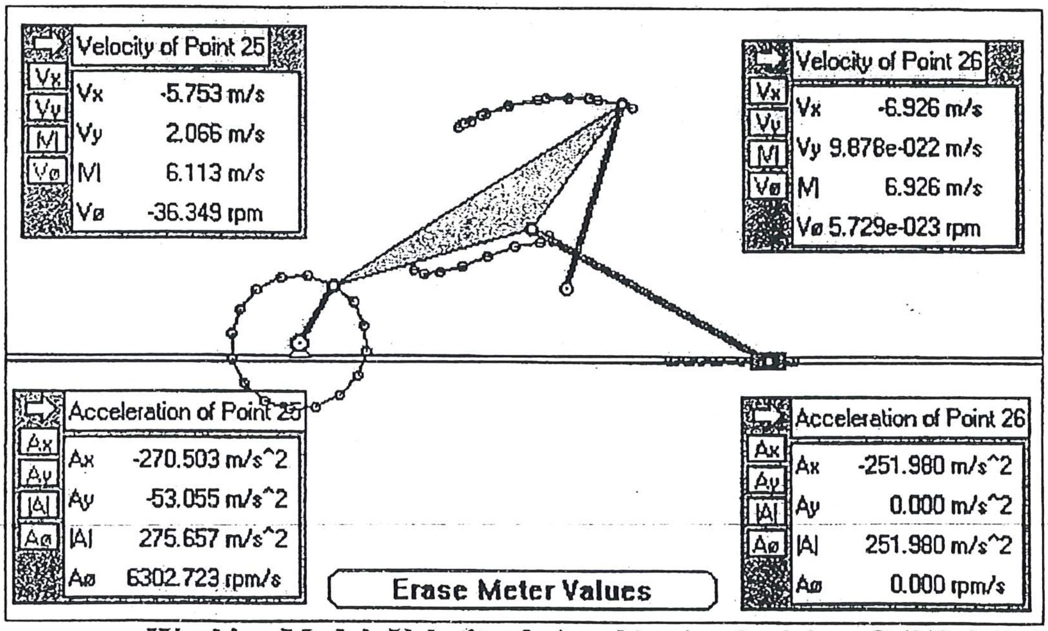
**Working Model: Plot of paths for joint A, B, C & D.**

# Velocity/Acceleration of Joints

In order to determine the correlation between the analytical and computational results for velocity at $\theta = 60^\circ$, a computational analysis was undertaken for the joints of the mechanism. The results obtained from the software are shown in figures :



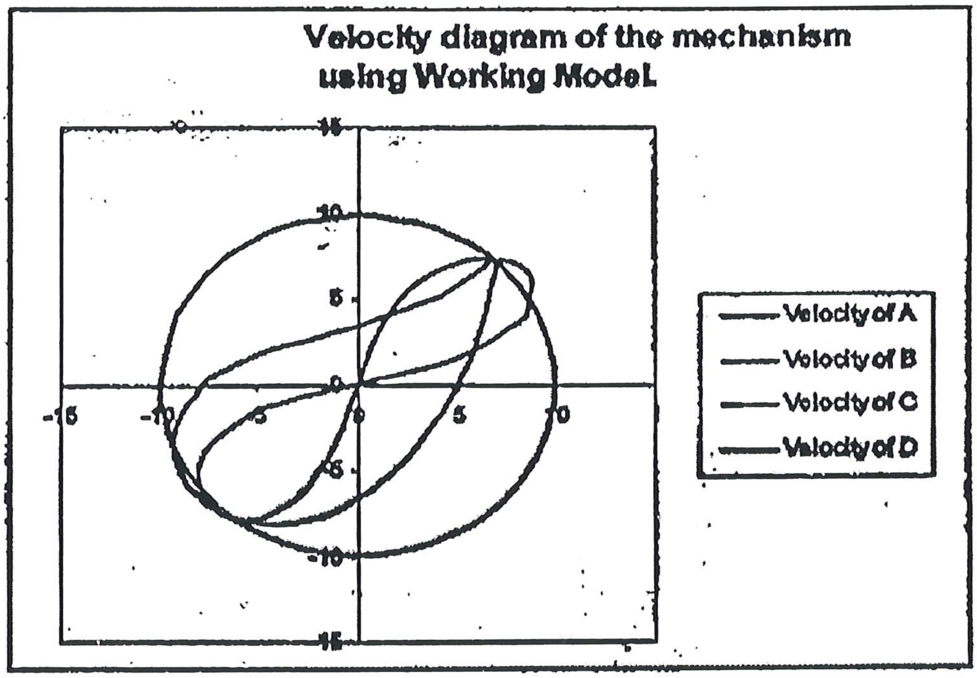

**Working Model: Velocity & Acceleration for joints A (28) & B (24).**

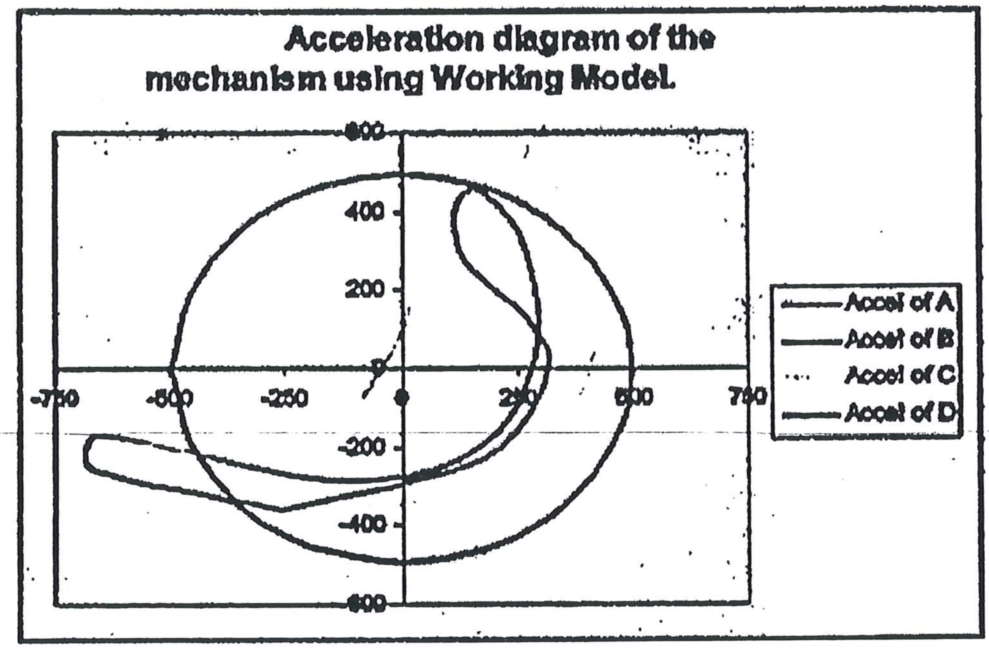



**Working Model: Velocity & Acceleration for joints C (25) & D (26).**

The results were obtained by running the mechanism for the velocity and acceleration at joints A, B, C and D. The results were exported from *'Working Model'* into Excel while the model was running.

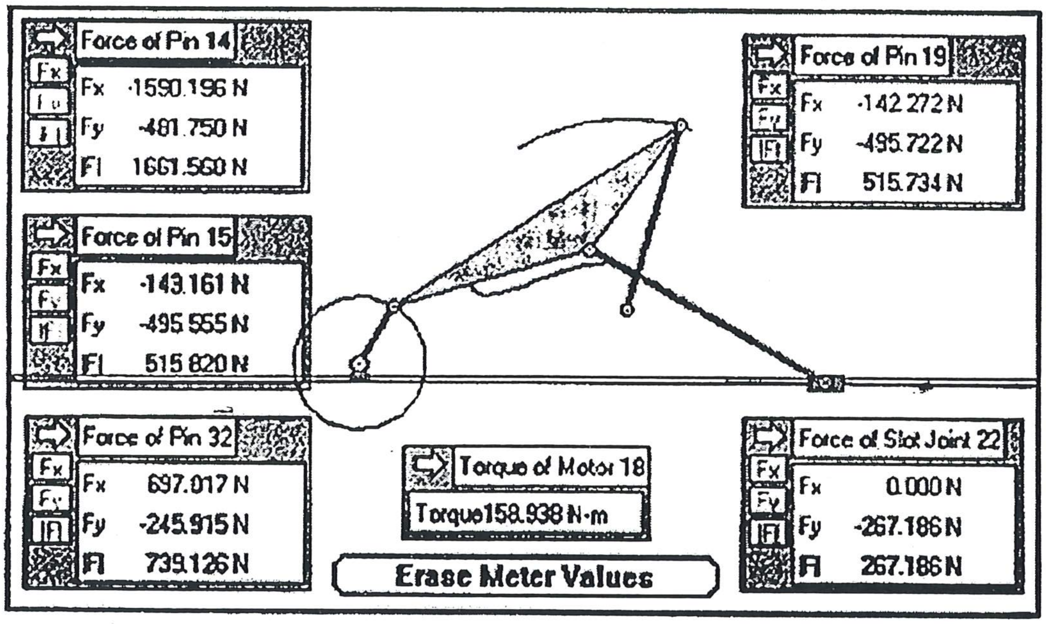
NOTE: The charts below were completed for one revolution of the crank (link OA).



## Force Analysis

In order to determine the forces present in the model, the forces at the linkages were investigated using the software program, with the result displayed in figure for $\theta = 60°$.




**Working Model: Forces results for joint A, B, C & D.**

The results of the force analysis using the matrix method and the software program, are displayed below

| Analytical Method | | Computational Method |
|---|---|---|
| Raw Data (N) | Joint Forces (N) | Software Generated (N) |
| $F_{12X}$ = -1990.307 | $F_{12}$ = 2227.07 | |
| $F_{12Y}$ = -999.273 | | |
| $F_{32X}$ = 1806.007 | $F_{32}$ = 1929.81 | $F_A$ = 1929.351 |
| $F_{32Y}$ = 680.073 | | |
| $F_{43X}$ = 335.819 | $F_{43}$ = 618.51 | $F_B$ = 747.243 |
| $F_{43Y}$ = 519.405 | | |
| $F_{53X}$ = 624.438 | $F_{53}$ = 656.18 | $F_C$ · 739.126 |
| $F_{53Y}$ = -201.632 | | |
| $F_{14X}$ = 4.139 | $F_{14}$ = 566.15 | $F_E$ = 709.716 |
| $F_{14Y}$ = 566.135 | | |
| $F_{65X}$ = 150.538 | $F_{65}$ = 295.15 | |
| $F_{65Y}$ = -253.872 | | |
| $F_{16Y}$ = -253.872 | $F_{16}$ = 265.05 | $F_F$ = 267.186 |
| $F_{16X}$ = 76.16 | | |
| $T_{12}$ = 143.491 Nm | | $T_{12}$ = 156.354 Nm |

**Force results for Matrix (Analytical) and computational Method.**

Finally, don't forget to provide a comprehensive summary of all results obtained using graphical, analytical and computational approach.

| | Graphical Method | | | Analytical Method | | Computational Method |
|---|---|---|---|---|---|---|
| Linear Velocity | Vector (m/s) | I.C (m/s) | θ (°) | Vector (m/s) | θ (°) | Vector (m/s) |
| $V_A$ | 7.1834 | 7.1834 | 150 | 7.183 | 150 | 7.183 |
| $V_B$ | 4.983 | 4.983 | 164 | 4.983 | 164 | 4.958 |
| $V_C$ | 6.103 | 6.103 | 160.2 | 6.103 | 160 | 6.113 |
| $V_D$ | 6.932 | 6.938 | 180 | 6.956 | 180 | 6.926 |

| Angular Velocity | Graphical Method: Vector (rad/s) | Analytical Method: Vector (rad/s) | Computational Method: Vector (rad/s) |
|---|---|---|---|
| $\omega_2$ | 51.31 | 51.31 | |
| $\omega_3$ | 3.7357 | 3.77045 | |
| $\omega_4$ | 12.4575 | 12.4569 | |
| $\omega_5$ | 4.2714 | 4.2823 | |

| Linear Acceleration | Graphical Method: Vector (m/s²) | Graphical Method: θ (°) | Analytical Method: Vector (m/s²) | Analytical Method: θ (°) | Computational Method: Vector (m/s) |
|---|---|---|---|---|---|
| $A_A$ | 368.58 | 240 | 368.58 | 240 | 368.635 |
| $A_B$ | 446.60 | 171.98 | 446.60 | 172 | 448.912 |
| $A_C$ | 280.598 | 195.84 | 280.598 | 196.8 | 275.657 |
| $A_D$ | 252.1 | 180 | 251.89 | 180 | 251.98 |

| Angular Acceleration | Graphical Method: Vector (rad/s²) | Analytical Method: Vector (rad/s²) | Computational Method: Vector (rad/s²) |
|---|---|---|---|
| $\alpha_A$ | 0 | 0 | |
| $\alpha_B$ | 600.6 | 600.5972 | |
| $\alpha_C$ | 1105.65 | 1105.646 | |
| $\alpha_D$ | 109.6 | 109.5921 | |

**Table of Velocity & Acceleration results for $\theta_2 = 60°$**

| Analytical Method | | Computational Method |
|---|---|---|
| Raw Data (N) | Joint Forces (N) | Software Generated (N) |
| $F_{12X} = -1990.307$ | $F_{12} = 2227.07$ | |
| $F_{12Y} = -999.273$ | | |
| $F_{32X} = 1806.007$ | $F_{32} = 1929.81$ | $F_A = 1929.351$ |
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| $F_{43Y} = 519.405$ | | |
| $F_{53X} = 624.438$ | $F_{53} = 656.18$ | $F_C = 739.126$ |
| $F_{53Y} = -201.632$ | | |
| $F_{14X} = 4.139$ | $F_{14} = 566.15$ | $F_E = 709.716$ |
| $F_{14Y} = 566.135$ | | |
| $F_{65X} = 150.538$ | $F_{65} = 295.15$ | |
| $F_{65Y} = -253.872$ | | |
| $F_{16Y} = -253.872$ | $F_{16} = 265.05$ | $F_S = 267.186$ |
| $F_{16X} = 76.16$ | | |
| $T_{12} = 143.491$ Nm | | $T_{12} = 156.354$ Nm |

**Table of Force results for $\theta_2 = 60^o$**

The stroke distance was determined using the graphical and analytical approach,

- **Graphical Method,**

Stroke Distance = **266.64** mm

- **Analytical Method,**

Stroke Distance = **266.1** mm

The shaking force, shaking torque and shaking moment at $\theta_2 = 60^o$, were found and are shown below respectively.

$F_S$ = **2173.84** N

$T_s$ = **-143.491** Nm

$M_S$ = **209.04** Nm

Plase note, that you are expected to propose and describe a method for balancing of this linkage mechanism based on the obtained values of shaking forces and moments.

| Analytical Method | | Computational Method |
|---|---|---|
| Raw Data (N) | Joint Forces (N) | Software Generated (N) |
| $F_{12X} = -1990.307$ | $F_{12} = 2227.07$ | |
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| $F_{32Y} = 680.073$ | | |
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| $F_{43Y} = 519.405$ | | |
| $F_{53X} = 624.438$ | $F_{53} = 656.18$ | $F_C = 739.126$ |
| $F_{53Y} = -201.632$ | | |
| $F_{14X} = 4.139$ | $F_{14} = 566.15$ | $F_E = 709.716$ |
| $F_{14Y} = 566.135$ | | |
| $F_{65X} = 150.538$ | $F_{65} = 295.15$ | |
| $F_{65Y} = -253.872$ | | |
| $F_{16Y} = -253.872$ | $F_{16} = 265.05$ | $F_S = 267.186$ |
| $F_{16X} = 76.16$ | | |
| $T_{12} = 143.491$ Nm | | $T_{12} = 156.354$ Nm |

**Table of Force results for $\theta_2 = 60^o$**

The stroke distance was determined using the graphical and analytical approach,

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- **Analytical Method,**

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$T_s$ = **-143.491** Nm

$M_S$ = **209.04** Nm

Plase note, that you are expected to propose and describe a method for balancing of this linkage mechanism based on the obtained values of shaking forces and moments.